Project: Worked examples
Title: Plate bending
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1. Scope and method

Bending of a thin, flat, isotropic plate under a lateral load by Kirchhoff small-deflection theory: circular plates (uniform pressure, central patch, ring load; clamped or simply supported) in closed form; rectangular plates simply supported (uniform pressure or central patch, Navier series) or clamped (Ritz, approximate). Maximum deflection, greatest von Mises stress at the surface, stress and deflection utilisations and the fundamental frequency.
Flexural rigidity: $D = E·t^3/(12(1 − ν^2))$; surface stress $σ = 6M/t^2$; von Mises $σ_{vm} = √(σ_x^2 − σ_x·σ_y + σ_y^2 + 3τ_{xy}^2)$ Circular, uniform q, clamped: $w = q·a^4/(64D)$; $σ_{edge} = 3q·a^2/(4t^2)$; simply supported: $w = (5 + ν)·q·a^4/(64(1 + ν)D)$; $σ_{centre} = 3(3 + ν)·q·a^2/(8t^2)$ Circular, central patch (radius c, pressure q) or ring (radius b, force F): $w = A + B·r^2 (+ q·r^4/64D)$ inside, $A′ + B′·r^2 + C′·\ln(r/a) + k·r^2·\ln(r/a)$ outside, $k = q·c^2/8D$ or $F/8πD$; constants from continuity of w, w′, M_r at the load radius and the edge conditions; $M_r = −D(w″ + ν·w′/r)$, $M_θ = −D(w′/r + ν·w″)$ Rectangular, simply supported: $w = Σ\, W_{mn}·\sin(mπx/a)·\sin(nπy/b)$ over odd m, n; $W_{mn} = q_{mn}/(π^4 D(m^2/a^2 + n^2/b^2)^2)$; $q_{mn} = 16q/(π^2 mn)$ (uniform) or $16q/(π^2 mn)·\sin(mπ/2)·\sin(nπ/2)·\sin(mπu/2a)·\sin(nπv/2b)$ (patch u × v); $M_x = Dπ^2 ΣW_{mn}(m^2/a^2 + ν·n^2/b^2)·\sin·\sin$, M_y likewise, $M_{xy} = −D(1 − ν)π^2 ΣW_{mn}(mn/ab)·\cos·\cos$; terms added until w, the centre moments and the corner M_{xy} each change by $< 10^{-6}$ over a doubling (cap 1024 × 1024; the corner sum of a uniform load, which converges like $1/N^2$, is Richardson-extrapolated before it is compared) Rectangular, clamped: Ritz, $w = Σ\, c_{ij}·X_i(x/a)·Y_j(y/b)$, $X_k(ξ) = ξ^2(1 − ξ)^2·P_{2k}(2ξ − 1)$; c from the minimum of the total potential energy; order K taken as 8, 12, 16, 20, 25, 30 until w and the stress change by $< 10^{-5}$ (or the change stops falling) Fundamental frequency: simply supported rectangle $ω_{11} = π^2(1/a^2 + 1/b^2)·√(D/(ρt))$, $f = ω/2π$; circular $ω = (λ^2/a^2)·√(D/(ρt))$, $λ^2$ from the Bessel characteristic equation; clamped rectangle by Rayleigh–Ritz with the same basis Checks: $σ_{vm} ≤ f_y/γ_{M0}$; $w_{max} ≤$ span/ratio (circular span = diameter, rectangle span = shorter side); WARN w/t > 0.5

Ref: Timoshenko & Woinowsky-Krieger, Theory of Plates and Shells Ch. 3 — Symmetrical bending of circular plates
Ref: Timoshenko & Woinowsky-Krieger, Theory of Plates and Shells Ch. 5 — Simply supported rectangular plates (Navier solution)
Ref: Timoshenko & Woinowsky-Krieger, Theory of Plates and Shells Ch. 6 — Rectangular plates with various edge conditions
Ref: Leissa, Vibration of Plates (NASA SP-160) Ch. 2, 4 — Natural frequencies of circular and rectangular plates
Ref: BS EN 1993-1-1:2005+A1:2014 6.1(1) — Partial factor γ_M0
Ref: BS EN 1993-1-1:2005+A1:2014 6.2.1(5) — Yield criterion for the cross-section

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
shape	Plate shape	rectangular		drawing: ENC-LID-01 rev A	any
support	Edge support, all round	simply supported		assumption: The lid sits on a gasket land and is bolted at close pitch: treated as simply supported, which is the safe side for deflection and the central stress.	any
a	Radius a (circular) or side a (rectangular)	300	mm	drawing: ENC-LID-01 rev A	10 … 10000 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
b	Side b (rectangular; not used for a circular plate)	200	mm	drawing: ENC-LID-01 rev A	10 … 10000 mm
t	Plate thickness	4	mm	drawing: ENC-LID-01 rev A	0.5 … 100 mm
load_type	Load	uniform		spec: Enclosure datasheet ENC-DS-07: internal pressure rise in a sealed enclosure on thermal cycling	any
load_basis	Uniform and patch loads given as	pressure q		assumption: Load given as a pressure.	any
q	Pressure q on the loaded area (uniform and patch loads, pressure basis)	20	kPa	spec: Enclosure datasheet ENC-DS-07: 0.2 bar differential	0.001 … 5000 kPa
P	Total force P (force basis: uniform, patch and ring loads; a ring load always uses P)	5	kN	default	0.001 … 10000 kN
r0	Patch radius (circular plate, central patch)	50	mm	default	1 … 5000 mm
u	Patch size in x (rectangular plate, central patch)	100	mm	default	1 … 10000 mm
v	Patch size in y (rectangular plate, central patch)	100	mm	default	1 … 10000 mm
r_ring	Ring radius (circular plate, ring load)	150	mm	default	1 … 5000 mm
E	Modulus of elasticity	70	GPa	reference: record:5eed0001-0000-4000-8000-000000000003 W3 generic seed 'Aluminium alloy (generic)', E: indicative, not from a standard; replace with the material record for the alloy on the drawing	0.1 … 500 GPa
nu	Poisson's ratio	0.33		reference: record:5eed0001-0000-4000-8000-000000000003 W3 generic seed 'Aluminium alloy (generic)', nu: indicative	0 … 0.45

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
rho	Density (for the frequency)	2700	kg/m^3	reference: record:5eed0001-0000-4000-8000-000000000003 W3 generic seed 'Aluminium alloy (generic)', rho: indicative	100 ... 20000 kg/m^3
f_y	Yield strength	150	MPa	assumption: Assumed 0.2 % proof strength for the sheet temper (the W3 generic seed has no strength); replace with the material record from the mill certificate.	5 ... 2000 MPa
gamma_M0	Partial factor γ_M0 on the resistance of the cross-section	1.1		assumption: EN 1999-1-1 partial factor γ_M1 = 1.1 for the resistance of cross-sections of aluminium, entered in the γ_M0 slot; to be checked against the standard and National Annex in use	1 ... 1.5
defl_ratio	Deflection limit: span/ratio (span = diameter of a circular plate, shorter side of a rectangle)	250		spec: Project specification: lid deflection limited to span/250 so the gasket stays seated	50 ... 2000

3. Assumptions

A1. Thin plate, small-deflection (Kirchhoff) theory: flat, isotropic, linear elastic, constant thickness; the middle surface is unstrained; shear deformation is neglected (span/t of 10 or more). (Scope of S9; a warning below span/t = 10.)
A2. The deflection stays small against the thickness: w/t above 0.5 means membrane action stiffens the plate and the small-deflection result is not valid (it overstates the deflection and the bending stress and ignores the membrane stress). (The sheet warns at w/t > 0.5.)
A3. The edge support is the same all round and ideal: a simply supported edge is held at zero deflection and free to rotate; a clamped edge allows no deflection or rotation. A simply supported rectangle has its corners held down (the corner reaction 2·M_xy is reported); if the corners can lift, the deflection and the central moments are larger than computed. (Navier solution; no elastic edges, no rotational springs.)
A4. The von Mises stress is that of the plane-stress field at the plate surface from the bending and twisting moments ($\sigma = 6M/t^2$); the through-thickness shear and the local stress under a concentrated load are not included. The stress governing a simply supported rectangle is at the centre or at a corner (twisting); a clamped rectangle is scanned on a 21 × 21 grid over a quarter of the plate. (Surface stresses; no local effects.)
A5. The rectangular patch and the circular patch are centred; a ring load is concentric. A load smaller than the thickness is a concentrated load whose local stress thin-plate theory does not give. (Warning

`patch-thin`; patches below 1 % (circular), 2 % (simply supported rectangle) or 25 % (clamped rectangle) of the plate size are refused.)
A6. The clamped rectangle is a Ritz approximation: the deflection converges quickly, the moments and stress more slowly (the sheet reports the order used and the last change; long plates and small patches converge slowest, and the last change understates the error). Compare a critical result with finite elements. (Approximate; the sheet states the order used and the last change.)
A7. Material properties are inputs taken from the W3 library (E, ν, ρ, f_y), each citing the material record, its revision and the page or table; the generic textbook seeds are indicative. (Record kind material.properties.)
A8. support (Edge support, all round) is taken as simply supported. (The lid sits on a gasket land and is bolted at close pitch: treated as simply supported, which is the safe side for deflection and the central stress.)
A9. load_basis (Uniform and patch loads given as) is taken as pressure q. (Load given as a pressure.)
A10. f_y (Yield strength) is taken as 150 MPa. (Assumed 0.2 % proof strength for the sheet temper (the W3 generic seed has no strength); replace with the material record from the mill certificate.)
A11. gamma_M0 (Partial factor γ_M0 on the resistance of the cross-section) is taken as 1.1. (EN 1999-1-1 partial factor γ_M1 = 1.1 for the resistance of cross-sections of aluminium, entered in the γ_M0 slot; to be checked against the standard and National Annex in use)

4. Working

[15] F_total = q·A_loaded — Kirchhoff theory
= 20 kPa × 60000 mm²
= 1.2 kN
Total load on the plate

[16] D = E*t^3/(12*(1 - nu^2)) — Kirchhoff theory
= 70 GPa × (4 mm)^3 / (12 × (1 - 0.33^2))
= 418.96 N·m
Flexural rigidity of the plate

[17] terms_used = odd terms per direction N (m, n = 1, 3, … 2N - 1) — T&WK Ch. 5
= 128 odd terms per direction (16384 terms), last change 6.5537e-7
= 128
Terms of the Navier series used

[18] w_max = 16q/(π⁶D)·Σ (-1)^((m+n)/2-1)/(m·n·(m²/a² + n²/b²)²), m, n odd — T&WK Ch. 5
= 128 odd terms per direction (16384 terms), last change 6.5537e-7
= 0.58996 mm
Maximum deflection (centre)

[19] M_x_c = D·π²·Σ W_mn(m²/a² + ν·n²/b²)·sin·sin at the centre — T&WK Ch. 5
= 128 odd terms per direction (16384 terms), last change 6.5537e-7
= 41.62 N·m/m
M_x at the centre

[20] M_y_c = D·π²·Σ W_mn(n²/b² + ν·m²/a²)·sin·sin at the centre — T&WK Ch. 5
= 128 odd terms per direction (16384 terms), last change 6.5537e-7
= 65.6 N·m/m
M_y at the centre

[21] M_xy_k = |D(1 - ν)·π²·Σ W_mn·(mn/ab)| at a corner, with the 1/N² tail of the sum removed
Richardson extrapolation (s_N + (s_N - s_N/2)/3) — T&WK Ch. 5
= 128 odd terms per direction (16384 terms), last change 6.5537e-7
= 32.874 N·m/m
Twisting moment at a corner

[22] R_corner = 2·M_xy at a corner — T&WK Ch. 5
= 2 × 32.874 N·m/m
= 0.065747 kN
Corner reaction holding the corner down

[23] M_vm_c = √(M_x² - M_x·M_y + M_y²) at the centre — von Mises
= M_x = 41.62, M_y = 65.6 N·m/m
= 57.492 N·m/m
Von Mises moment at the centre

[24] M_vm_k = √3·|M_xy| at a corner — von Mises
= M_xy = 32.874 N·m/m
= 56.939 N·m/m
Von Mises moment at a corner (pure twist)

[25] omega_11 = π²(1/a² + 1/b²)·√(D/(□t)) — T&WK Ch. 5
= a = 300 mm, b = 200 mm, D = 418.96 N·m, □t = 10.8 kg/m²
= 2219.8 rad/s
Angular frequency of the (1,1) mode

[26] M_vm = √(M_1² - M_1·M_2 + M_2² + 3M_12²) at the governing point — von Mises
= centre
= 57.492 N·m/m
Greatest von Mises moment per unit width

[27] sigma_max = 6·M_vm/t² — Kirchhoff theory
= 6 × 57.492 N·m/m / (4 mm)²
= 21.56 MPa
Greatest von Mises surface stress

[28] sigma_principal = 6·M_principal/t² — Kirchhoff theory
= 6 × 65.6 N·m/m / (4 mm)²
= 24.6 MPa
Greatest principal surface stress

[29] f1 = □/2π — Leissa
= D = 418.96 N·m, □t = 10.8 kg/m²
= 353.29 Hz
Fundamental natural frequency (ω is the angular frequency, rad/s; f is in Hz)

[30] w_over_t = w_max/t — Kirchhoff theory
= 0.58996 mm / 4 mm
= 0.14749
Deflection over thickness

[31] u_stress = sigma_max/(f_y/gamma_M0) — BS EN 1993-1-1 6.2.1(5)
= 21.56 MPa / (150 MPa / 1.1)
= 0.1581
Stress utilisation

[32] w_lim = span/ratio — Serviceability
= 200 mm / 250
= 0.8 mm
Deflection limit

[33] u_defl = w_max/w_lim — Serviceability
= 0.58996 mm / 0.8 mm
= 0.73745
Deflection utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
w_max	0.58996 mm	Maximum deflection
sigma_max	21.56 MPa	Greatest von Mises stress
sigma_principal	24.6 MPa	Greatest principal stress
f1	353.29 Hz	Fundamental frequency
u_stress	0.1581	Stress utilisation

SYMBOL	VALUE	DESCRIPTION
u_defl	0.73745	Deflection utilisation
w_over_t	0.14749	Deflection over thickness
D	418.96 N·m	Flexural rigidity
terms_used	128	Series terms used
M_x_c	41.62 N·m/m	M_x at the centre
M_y_c	65.6 N·m/m	M_y at the centre
M_xy_k	32.874 N·m/m	Corner twisting moment
R_corner	0.065747 kN	Corner reaction
location	"centre"	Governing stress point

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Deflection w_max ≤ span/ratio	u_defl = 0.73745	u_defl ≤ 1	0.737	Serviceability
PASS	Stress σ_vm ≤ f_y/γ_M0	u_stress = 0.1581	u_stress ≤ 1	0.158	6.2.1(5)

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: f_y (Material properties); gamma_M0 (EN 1993 National Annex factors); defl_ratio (Deflection limit). Enter each as one so its source, edition and licence travel with it.
WARNING — input.P: P (Total force P (force basis: uniform, patch and ring loads; a ring load always uses P)) was not entered; the template default 5 kN is used.
WARNING — input.r0: r0 (Patch radius (circular plate, central patch)) was not entered; the template default 50 mm is used.
WARNING — input.u: u (Patch size in x (rectangular plate, central patch)) was not entered; the template default 100 mm is used.
WARNING — input.v: v (Patch size in y (rectangular plate, central patch)) was not entered; the template default 100 mm is used.
WARNING — input.r_ring: r_ring (Ring radius (circular plate, ring load)) was not entered; the template default 150 mm is used.

8. Verification

Template struct.plate-bending@1.2.1 Content hash 7776225ec2491f68 Engine calc-engine/1.0.0
Known-answer cases passed 11/11 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0